

Simulation of mosquito-borne disease spread in Denmark

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Objective

Build a generic model to simulate the spatial spread of mosquito-borne disease in Denmark

Preparedness framework

Developed for a disease-free setting, the model is designed as a preparedness tool with:

- A flexible host structure (livestock and wildlife, e.g. wild birds)
- Multiple transmission pathways (mosquito-borne and host-to-host)

We want to be able to test control measures and identify main drivers of transmission

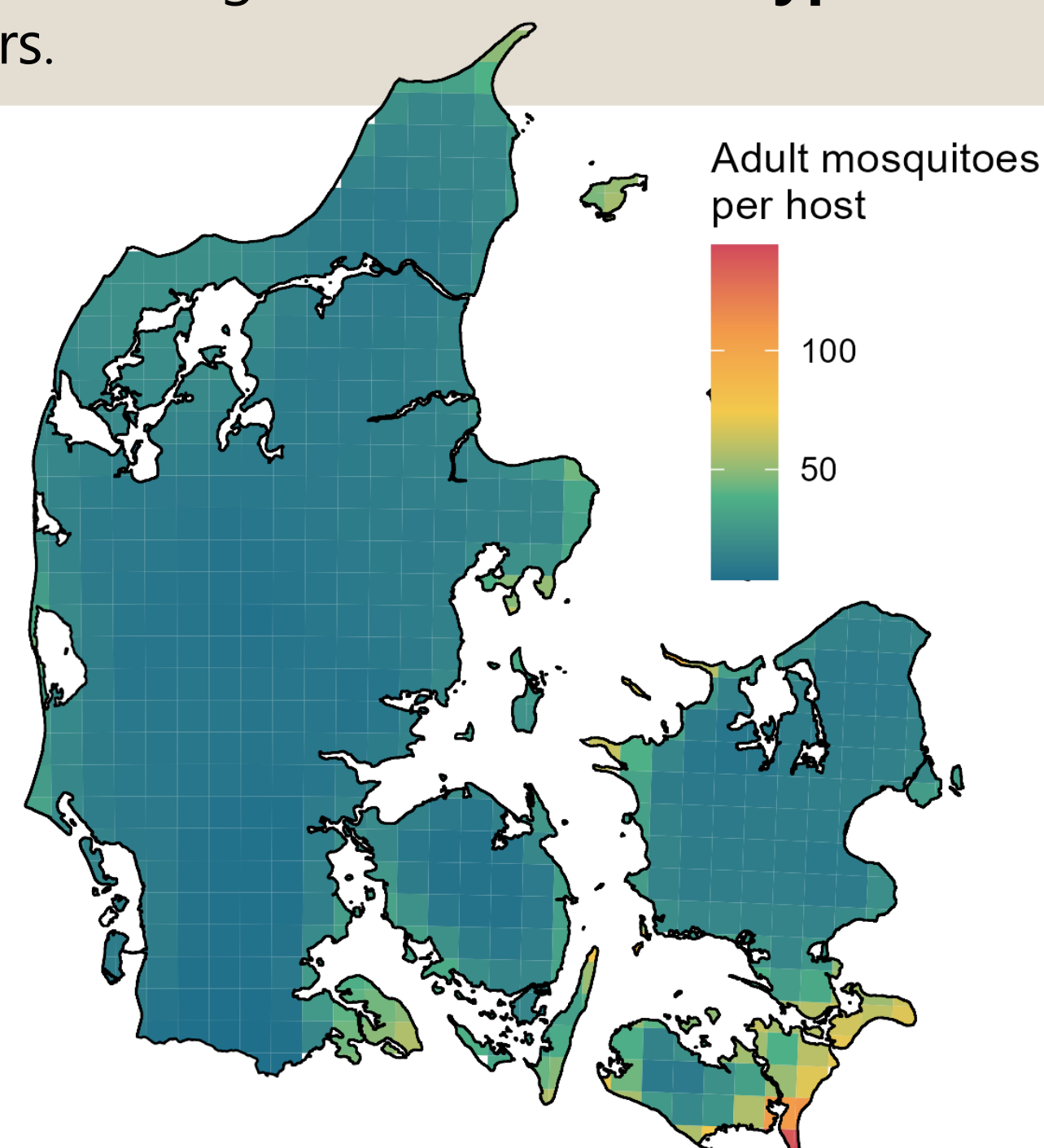
What we already have

1. Mosquito abundance predictions

In a parallel project we are developing a model predicts **abundance of mosquitoes** at different live stages based on **temperature** (See Figure 1).

→ Work in progress: adding **rainfall** and **land type** as additional drivers.

Figure 1. Number of Adult mosquitoes per host on the 2nd of September 2024. Predicted by a population dynamic model with temperature dependence. Temperature has been obtained in a 10 x 10 km grid cell resolution.

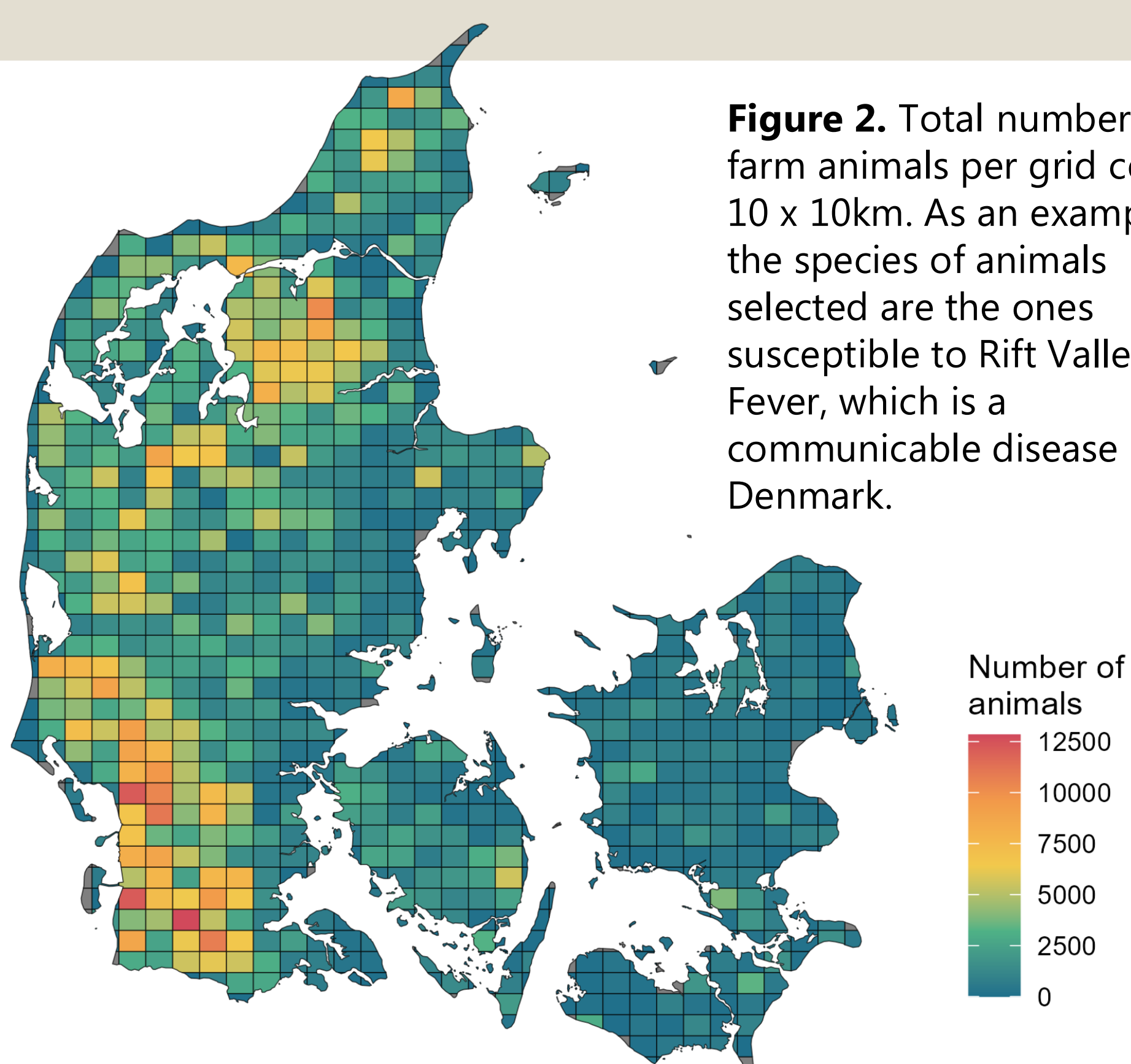


2. Farm data

- **Location** of farms
- **Number of animals** on each farm (over time)
- **Species** and number of each species (over time)

→ Work in progress: Acquiring **animal movement data**

Figure 2. Total number of farm animals per grid cell of 10 x 10km. As an example, the species of animals selected are the ones susceptible to Rift Valley Fever, which is a communicable disease in Denmark.



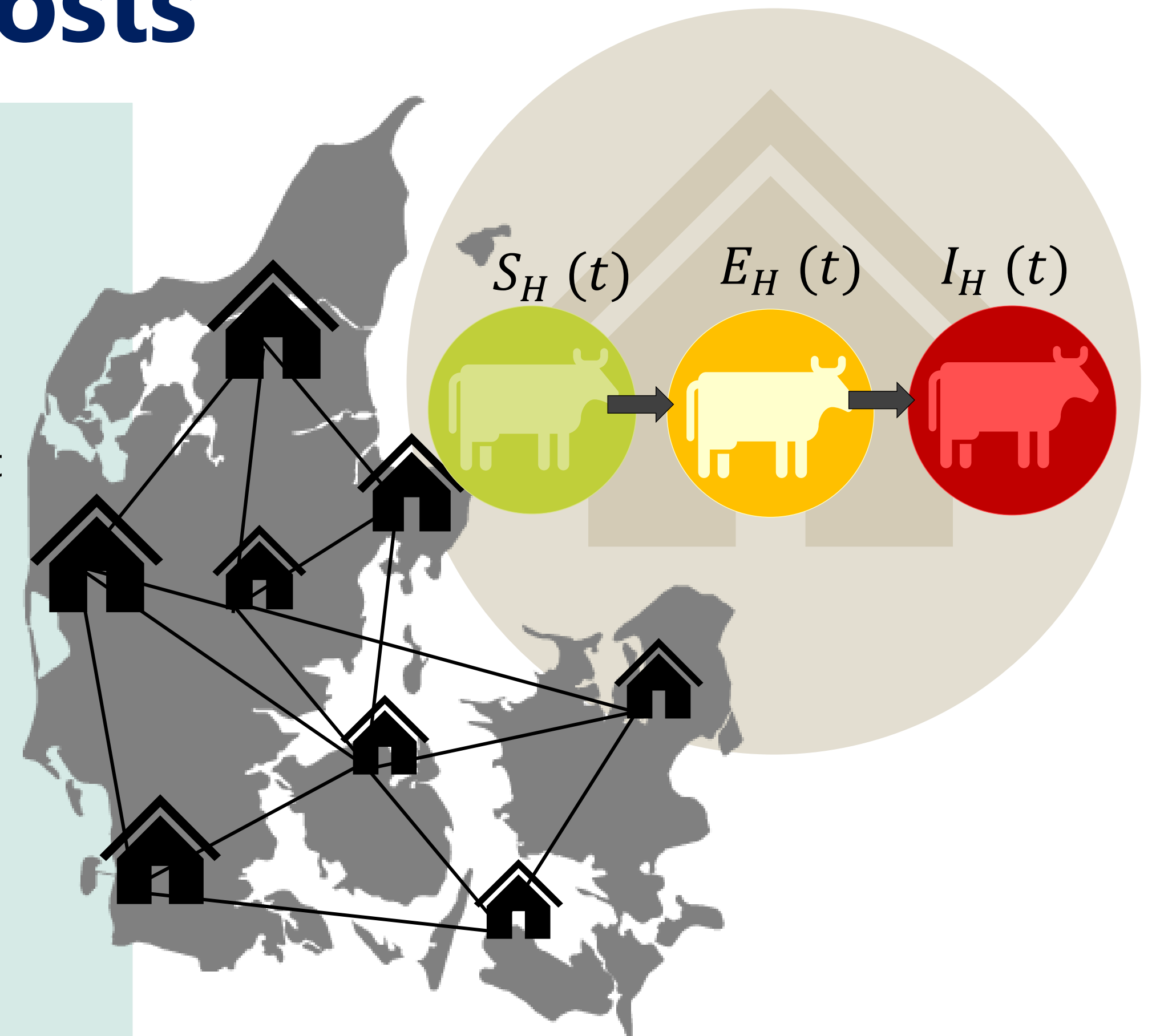
Modelling the hosts

Compartmental mechanistic model (SEIR)

- **States:** Susceptible – Exposed – Infected – Recovered, for each species in each farm
- **Exposed compartment:** Contains multiple substages to better represent the latency period
- **Stochastic:** Accounts for individual differences between hosts

Advantages:

- Can incorporate **between-host transmission**
- **Control measures** (e.g., vaccination) are easily applied
- **Animal movement data** can be included to simulate disease spread



Modelling pathogen spread via mosquitoes – *In progress*

Farm-specific mechanistic model

- **Spatial spread:** Virus spreads via mosquito movement between farms.
- **Local dynamics:** SEI mosquito populations modelled at the farm level.

Open question: How should mosquito movement between farms be represented?

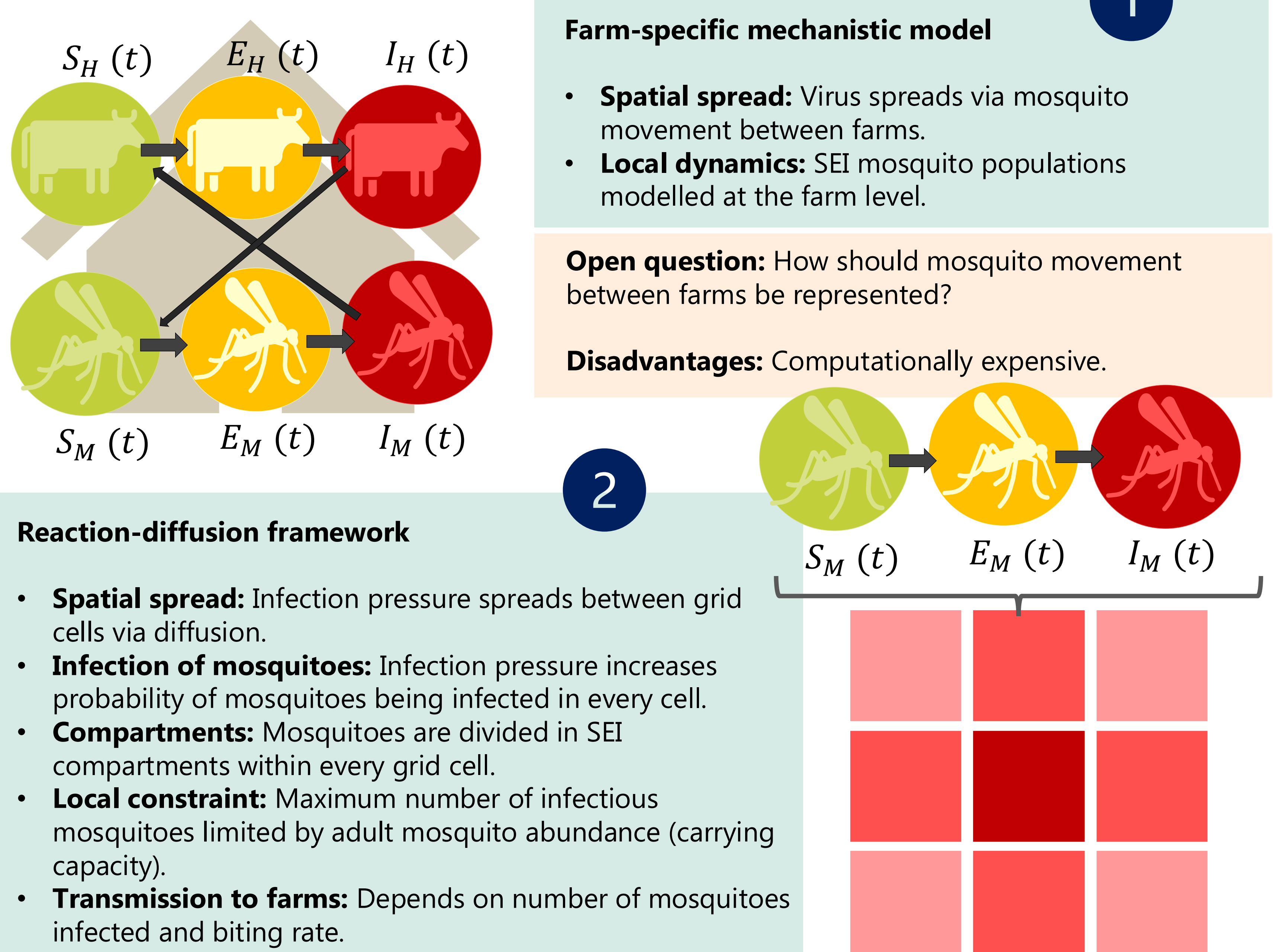
Disadvantages: Computationally expensive.

Reaction-diffusion framework

- **Spatial spread:** Infection pressure spreads between grid cells via diffusion.
- **Infection of mosquitoes:** Infection pressure increases probability of mosquitoes being infected in every cell.
- **Compartments:** Mosquitoes are divided in SEI compartments within every grid cell.
- **Local constraint:** Maximum number of infectious mosquitoes limited by adult mosquito abundance (carrying capacity).
- **Transmission to farms:** Depends on number of mosquitoes infected and biting rate.

Open question: How fast should the pathogen spread spatially?

Disadvantages: Mosquito population may become overly well-mixed depending on grid cell size.



References

- (1) Mordecai EA, Caldwell JM, Grossman MK, Lippi CA, Johnson LR, Neira M, et al. Thermal biology of mosquito-borne disease. *Ecol Lett*. 2019;22(10):1690–708.

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